Digital Signal Processing

1. a) i) Describe in one or two sentences the key features of Discrete Fourier Series (DFS) analysis and its relationship to the Discrete Fourier Transform (DFT). Include relevant formulae. [2]

ii) Consider two periodic discrete-time signals $x_{1p}(n)$ and $x_{2p}(n)$, both with period N. The corresponding DFS coefficients are given by $X_{1p}(k)$ and $X_{2p}(k)$ respectively. Let

$$X_{3p}(k) = X_{1p}(k)X_{2p}(k).$$

Derive an expression for $x_{3p}(n)$, as the inverse DFS of $X_{3p}(k)$, in terms of x_{1p} and x_{2p}.

[4]

b) i) Give a detailed description of linear convolution and circular convolution in discrete time and state the difference(s) between these two operations. Also state in which circumstances each would be used. [4]

ii) Showing clearly the steps taken, compute the linear and circular convolution of the two discrete-time sequences

$$p(n) = [0\ -2\ 1\ 3] \text{ and } q(n) = [5\ -1\ -2\ 0].$$

[2]

iii) First describe a technique by which circular convolution can be used to compute linear convolution, with the aid of any relevant diagrams. Then use the technique described for the sequences $p(n)$ and $q(n)$ from part ii). [4]

iv) Consider an LTI system with impulse response $h(l)$, input signal $x(l)$ and output signal $y(l)$.

Derive an expression, involving the convolution operation, for the cross-correlation between the input signal and the output signal.

Consider the case in which the input signal and output signal are known but the impulse response $h(l)$ is unknown. Develop and describe a method to measure $h(l)$ in this case. State any required assumptions. [4]

2. a) Consider a discrete-time signal

$$x(n), \quad n = 0,\ 1,\ \ldots,\ N-1$$

and its DFT denoted $X(k)$.
Next consider a signal derived from $x(n)$ given by

$$x_p(n) = \begin{cases} x(n), & n = 0,\ 1,\ \ldots,\ N-1 \\ 0, & n = N,\ N+1,\ \ldots,\ M-1 \end{cases}$$

and its DFT denoted $X_p(k)$.

i) Sketch the signals $x(n)$ and $x_p(n)$ for $x(n) = n(-1)^n$ with $N = 5$ and $M = 7$. [2]

ii) Write down the inverse DFT expression giving $x_p(n)$ in terms of $X_p(k)$. [2]

iii) Derive the relationship between $X_p(k)$ and $x(n)$. [2]

iv) Now consider the case with $M = NL$ for integer L. State and explain the differences and similarities between the values of $X(k)$ and $X_p(k)$. [3]

v) An example signal $x(n)$ is shown on the left in Fig. 2.1. The modulus of $X(k)$ is shown on the right. It is proposed to compute the DFT of $x(n)$ by first forming a new signal $x_p(n)$, derived from $x(n)$, and then using an 8-point FFT. Draw a sketch of $|X_p(k)|$. [3]

vi) Explain how you would find the magnitude spectrum of $X(k)$ at $\omega = 4\pi/7$ using the result of the FFT in part v). [3]

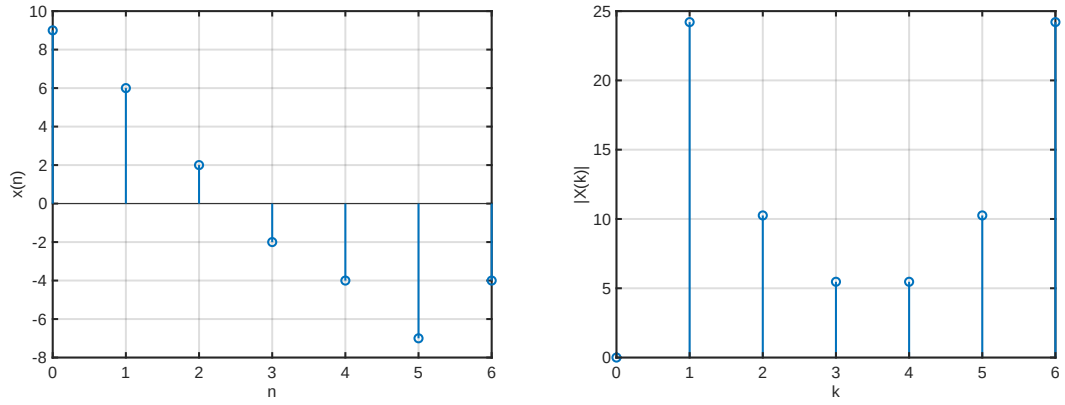


Figure 2.1

b) A finite impulse response filter with input signal $x(n)$ and output signal $y(n)$ has a transfer function

$$H(z) = 1 - 1.3435z^{-1} + 0.9025z^{-2}.$$

i) Draw a labelled block diagram of the direct form implementation of this filter.
Write down the difference equation for $y(n)$. [2]

ii) The two-stage lattice implementation of this filter is shown in the block diagram of Fig. 2.2. Write expressions, in terms of the input signal, for $y_2(n)$ and $w_2(n)$. Hence, find the values of k_1 and k_2. [3]

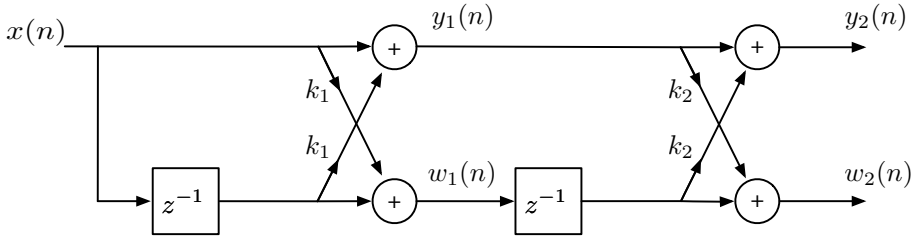


Figure 2.2

3. a) Consider a time index n over the range $n = 0,\ 1,\ \ldots,\ N-1$.

For a real discrete-time signal $x(n)$, also consider another signal

$$y(n) = y(2N-1-n) = x(n).$$

Derive an expression for the $2N$-point DFT of $y(n)$ involving a sum of cosine functions. [7]

b) i) Using appropriate block diagrams, state and explain the Noble Identities. Explain how the Noble Identities can be advantageously applied in the design of multirate signal processing systems. [4]

ii) It is desired to downsample a signal $x(n)$ by a factor of 3. An FIR filter $H(z)$ with a lowpass frequency response is to be used in the downsampling system. The impulse response of the filter is denoted $h(i)$ for $i = 0, 1, \ldots, 8$.

Derive a Type I polyphase implementation of $H(z)$ and sketch a block diagram of the polyphase filter implementation. [4]

iii) Consider a discrete-time signal

$$x(n) = [1, 3, 2, 3, 2, 1, 0]$$

and a filter with impulse response

$$h(n) = [0, 0, 0.08, 0.25, 0.35, 0.25, 0.08, 0, 0].$$

In order to show the use of the downsampling structure of part ii) including a downsampling factor of 3 and the Type I polyphase filter implementation:

- compute the sample values of the intermediate signals at the inputs and at the outputs of all polyphase filtering components; [3]
- determine the sample values of the downsampled output signal. [2]

Show your working.

4. a) i) In the context of LTI systems, explain the relationship between the s-domain and the z-domain using appropriate diagrams. [2]

ii) Sketch on both your diagrams the loci corresponding to

- s being purely imaginary,
- s having equal real and imaginary parts. [3]

b) Prove that the z-transform of $nx(n)$ is given by $\mathrm{Z}\{nx(n)\} = -z\frac{dX(z)}{dz}$. [3]

c) Prove that the z-transform of $b^{-n}x(n)$ is given by $Z\{b^{-n}x(n)\} = X(bz)$. [2]

d) Consider the sequences

$$X(z) = a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + \dots$$

and

$$Y(z) = a_0 z^2 - a_1 b^1 z^3 - 2a_2 b^2 z^4 - 3a_3 b^3 z^5 - \dots \ .$$

Find and simplify an expression for $Y(z)$ in terms of $\frac{dX(z)}{dz}$. [10]